Addition of Angular Momenta and Clebsch–Gordan Coefficients

A Quick Review with Three Teaching Examples

Following the treatments of Griffiths (*Intro. to Elementary Particles*) and Zettili (*Quantum Mechanics: Concepts and Applications*)

Quick Review: The Core Ideas

The Problem

Suppose we have two angular momentum operators $\hat{\mathbf{J}}_1$ and $\hat{\mathbf{J}}_2$ (these could be orbital angular momenta, spins, or one of each). The **total** angular momentum is

$$\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2.$$

We know the eigenstates of each subsystem individually:

$$\hat{J}_1^2 \left|j_1, m_1\right\rangle = \hbar^2 j_1(j_1+1) \left|j_1, m_1\right\rangle, \qquad \hat{J}_{1z} \left|j_1, m_1\right\rangle = \hbar\, m_1 \left|j_1, m_1\right\rangle,$$

and likewise for $\hat{\mathbf{J}}_2$. The **uncoupled basis** is the tensor product:

$$\left|j_1, m_1\right\rangle \otimes \left|j_2, m_2\right\rangle \equiv \left|j_1, m_1;\, j_2, m_2\right\rangle.$$

We want to re-express these states in the **coupled basis** $\left|j, m\right\rangle$, eigenstates of $\hat{J}^2$ and $\hat{J}_z$.

Key Rules

Rules for Addition of Angular Momenta

1. The total quantum number j ranges over:

$$j = |j_1 - j_2|,\ |j_1 - j_2| + 1,\ \ldots,\ j_1 + j_2.$$

2. The magnetic quantum number is always additive:

$$m = m_1 + m_2.$$

3. The **completeness relation** (change of basis):

$$\boxed{|j, m\rangle = \sum_{m_1+m_2=m} \langle j_1, m_1;\ j_2, m_2 | j, m\rangle\ |j_1, m_1;\ j_2, m_2\rangle\,.}$$

   The expansion coefficients $\langle j_1, m_1;\ j_2, m_2 | j, m\rangle$ are the **Clebsch–Gordan (CG) coefficients**.

4. Dimension check: the total number of states is preserved:

$$(2j_1 + 1)(2j_2 + 1) = \sum_{j=|j_1-j_2|}^{j_1+j_2} (2j + 1).$$

How to Build the Coupled States (Ladder Method)

The standard procedure (used in both Griffiths and Zettili) is:

1. **Start at the top.** The *stretched state* $|j_{\max}, j_{\max}\rangle$ with $j_{\max} = j_1 + j_2$ is unique:

$$|j_1 + j_2,\ j_1 + j_2\rangle = |j_1, j_1;\ j_2, j_2\rangle\,.$$

2. **Apply the lowering operator** $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$ repeatedly using

$$\hat{J}_-\,|j, m\rangle = \hbar\sqrt{j(j+1) - m(m-1)}\ \ |j, m-1\rangle$$

   to generate all $|j_{\max}, m\rangle$ states.

3. **Use orthogonality** to find $|j_{\max} - 1,\ j_{\max} - 1\rangle$: it must be orthogonal to $|j_{\max},\ j_{\max} - 1\rangle$ within the $m = j_{\max} - 1$ subspace.

4. Repeat steps 2–3 for each successively lower value of j.

Example 1: Two Spin-$\frac{1}{2}$ Particles $\left(\frac{1}{2} \otimes \frac{1}{2}\right)$(1/2 x 1/2)

Example 1 — Adding Two Spin-$\frac{1}{2}$ Particles

This is the most fundamental example—the hydrogen atom's electron and proton spins, two-electron systems, or quark–antiquark spin structure. Griffiths uses this case extensively in his particle physics text.

Setup. We have $j_1 = \frac{1}{2}$, $j_2 = \frac{1}{2}$. Each can be $m = +\frac{1}{2}$ ("↑") or $m = -\frac{1}{2}$ ("↓").

Allowed j values: $j = |j_1 - j_2|$ to $j_1 + j_2 \Rightarrow j = 0,\ 1$.
Dimension check: $(2)(2) = 4$ and $(2 \cdot 1 + 1) + (2 \cdot 0 + 1) = 3 + 1 = 4$. ✓

Step 1: Stretched state.

$$|1, 1\rangle = |\uparrow\uparrow\rangle \equiv |\tfrac{1}{2}, \tfrac{1}{2};\ \tfrac{1}{2}, \tfrac{1}{2}\rangle\,.$$

Step 2: Apply $\hat{J}_-$ to get $|1, 0\rangle$.
On the left: $\hat{J}_-\,|1, 1\rangle = \hbar\sqrt{1(2) - 1(0)}\ |1, 0\rangle = \hbar\sqrt{2}\ |1, 0\rangle$.
On the right: $(\hat{J}_{1-} + \hat{J}_{2-})\,|\uparrow\uparrow\rangle = \hbar\,|\downarrow\uparrow\rangle + \hbar\,|\uparrow\downarrow\rangle$.
Equating:

$$\boxed{|1, 0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big).}$$

Step 3: Apply $\hat{J}_-$ again to get $|1, -1\rangle$.

$$|1, -1\rangle = |\downarrow\downarrow\rangle\,.$$

Step 4: Orthogonality gives the singlet $|0, 0\rangle$.
In the $m = 0$ subspace, the state orthogonal to $|1, 0\rangle$ is:

$$\boxed{|0, 0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big).}$$

Summary: $\frac{1}{2} \otimes \frac{1}{2}$

Coupled state $\lvert j, m\rangle$	Uncoupled expansion	Name
$\lvert 1, +1\rangle$	$\lvert\uparrow\uparrow\rangle$	3*Triplet ($s = 1$)
$\lvert 1,\ 0\rangle$	$\frac{1}{\sqrt{2}}\big(\lvert\uparrow\downarrow\rangle + \lvert\downarrow\uparrow\rangle\big)$	
$\lvert 1, -1\rangle$	$\lvert\downarrow\downarrow\rangle$	
$\lvert 0,\ 0\rangle$	$\frac{1}{\sqrt{2}}\big(\lvert\uparrow\downarrow\rangle - \lvert\downarrow\uparrow\rangle\big)$	Singlet ($s = 0$)

Particle physics context (Griffiths). The pion (π) is a spin-0 bound state of a quark and antiquark — it lives in the *singlet*. The rho meson (ρ) is the spin-1 partner — it lives in the *triplet*. The CG decomposition here is the starting point for understanding meson spectroscopy.

Example 2: Spin-1 ⊗x Spin-$\frac{1}{2}$ $\left(1 \otimes \frac{1}{2}\right)$(1 x 1/2)

Example 2 — Adding j_1

This case arises when coupling orbital angular momentum $\ell = 1$ with spin $s = \frac{1}{2}$ (e.g., the p-orbital electron in hydrogen), or when adding the isospin of a pion ($I = 1$) with a nucleon ($I = \frac{1}{2}$) as Griffiths discusses for the Δ resonance.

Setup. $j_1 = 1$, $j_2 = \frac{1}{2}$. Allowed j: $j = \frac{1}{2}, \frac{3}{2}$.
Dimension check: $(3)(2) = 6$ and $(2 \cdot \frac{3}{2} + 1) + (2 \cdot \frac{1}{2} + 1) = 4 + 2 = 6$. ✓

Step 1: Stretched state.

$$|\tfrac{3}{2}, \tfrac{3}{2}\rangle = |1, 1;\ \tfrac{1}{2}, \tfrac{1}{2}\rangle \equiv |1, 1\rangle\, |\uparrow\rangle\,.$$

Step 2: Lower to get $|\frac{3}{2}, \frac{1}{2}\rangle$.

Applying $\hat{J}_-$ to the left: $\hat{J}_-\, |\frac{3}{2}, \frac{3}{2}\rangle = \hbar\sqrt{3}\ |\frac{3}{2}, \frac{1}{2}\rangle\,.$

Applying $\hat{J}_{1-} + \hat{J}_{2-}$ to the right:

$$\hat{J}_{1-}\, |1, 1\rangle\, |\uparrow\rangle + |1, 1\rangle\, \hat{J}_{2-}\, |\uparrow\rangle = \hbar\sqrt{2}\ |1, 0\rangle\, |\uparrow\rangle + \hbar\ |1, 1\rangle\, |\downarrow\rangle\,.$$

Equating:

$$\boxed{|\tfrac{3}{2}, \tfrac{1}{2}\rangle = \sqrt{\tfrac{2}{3}}\ |1, 0\rangle\, |\uparrow\rangle + \sqrt{\tfrac{1}{3}}\ |1, 1\rangle\, |\downarrow\rangle\,.}$$

The CG coefficients here are: $\langle 1, 0;\ \frac{1}{2}, \frac{1}{2}\,|\,\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}}$ and $\langle 1, 1;\ \frac{1}{2}, -\frac{1}{2}\,|\,\frac{3}{2}, \frac{1}{2}\rangle = \sqrt{\frac{1}{3}}$.

Step 3: Orthogonality for $|\frac{1}{2}, \frac{1}{2}\rangle$.

The state orthogonal to $|\frac{3}{2}, \frac{1}{2}\rangle$ in the $m = \frac{1}{2}$ subspace is:

$$\boxed{|\tfrac{1}{2}, \tfrac{1}{2}\rangle = \sqrt{\tfrac{1}{3}}\ |1, 0\rangle\, |\uparrow\rangle - \sqrt{\tfrac{2}{3}}\ |1, 1\rangle\, |\downarrow\rangle\,.}$$

Continuing the ladder one obtains all six states. The complete table is:

$\lvert j, m\rangle$	Expansion in $\lvert m_1\rangle\, \lvert m_2\rangle$
$\lvert\frac{3}{2}, +\frac{3}{2}\rangle$	$\lvert 1, 1\rangle\, \lvert\uparrow\rangle$
$\lvert\frac{3}{2}, +\frac{1}{2}\rangle$	$\sqrt{\frac{2}{3}}\, \lvert 1, 0\rangle\, \lvert\uparrow\rangle + \sqrt{\frac{1}{3}}\, \lvert 1, 1\rangle\, \lvert\downarrow\rangle$
$\lvert\frac{3}{2}, -\frac{1}{2}\rangle$	$\sqrt{\frac{1}{3}}\, \lvert 1, -1\rangle\, \lvert\uparrow\rangle + \sqrt{\frac{2}{3}}\, \lvert 1, 0\rangle\, \lvert\downarrow\rangle$
$\lvert\frac{3}{2}, -\frac{3}{2}\rangle$	$\lvert 1, -1\rangle\, \lvert\downarrow\rangle$
$\lvert\frac{1}{2}, +\frac{1}{2}\rangle$	$\sqrt{\frac{1}{3}}\, \lvert 1, 0\rangle\, \lvert\uparrow\rangle - \sqrt{\frac{2}{3}}\, \lvert 1, 1\rangle\, \lvert\downarrow\rangle$
$\lvert\frac{1}{2}, -\frac{1}{2}\rangle$	$\sqrt{\frac{2}{3}}\, \lvert 1, -1\rangle\, \lvert\uparrow\rangle - \sqrt{\frac{1}{3}}\, \lvert 1, 0\rangle\, \lvert\downarrow\rangle$

Physics context. In atomic physics (Zettili), this is spin-orbit coupling for a p-electron, giving rise to the $j = \frac{3}{2}$ and $j = \frac{1}{2}$ fine-structure levels. In particle physics (Griffiths), this same algebra governs pion–nucleon scattering: $\pi N \to \Delta$. The Δ^{++} baryon resonance sits in the $I = \frac{3}{2}$ isospin multiplet, and the CG coefficients directly determine the relative cross-section ratios for $\pi^+ p$, $\pi^0 p$, $\pi^- p$ scattering.

Example 3: Two Spin-1 Particles $(1 \otimes 1)$(1 x 1)

Example 3 — Adding j_1

This is a richer example that illustrates the full power of the CG machinery. It arises in deuteron physics (coupling two nucleon spins in the $S = 1$ channel with $L = 1$ orbital angular momentum) and in the classification of light mesons from $q\bar{q}$ systems with $L = 1$.

Setup. $j_1 = 1$, $j_2 = 1$.
Allowed j values: $j = 0,\ 1,\ 2$.
Dimension check: $(3)(3) = 9$ and $(5) + (3) + (1) = 9$. ✓

Let us use the shorthand $|m_1, m_2\rangle$ for $|1, m_1;\ 1, m_2\rangle$.

Step 1: Stretched state.

$$|2, 2\rangle = |1, 1\rangle\,.$$

Step 2: Lower to $|2, 1\rangle$.
$\hat{J}_-\,|2, 2\rangle = 2\hbar\,|2, 1\rangle$. $\quad(\hat{J}_{1-} + \hat{J}_{2-})\,|1, 1\rangle = \hbar\sqrt{2}\;|0, 1\rangle + \hbar\sqrt{2}\;|1, 0\rangle$.

$$\boxed{|2, 1\rangle = \frac{1}{\sqrt{2}}\big(|1, 0\rangle + |0, 1\rangle\big).}$$

Step 3: Lower again to $|2, 0\rangle$.
Applying $\hat{J}_-$ to $|2, 1\rangle$: LHS $= \hbar\sqrt{6}\;|2, 0\rangle$.
RHS: Apply $\hat{J}_{1-} + \hat{J}_{2-}$ to $\frac{1}{\sqrt{2}}(|1, 0\rangle + |0, 1\rangle)$:

$$\frac{1}{\sqrt{2}}\big(\sqrt{2}\hbar\,|0, 0\rangle + \sqrt{2}\hbar\,|1, -1\rangle + \sqrt{2}\hbar\,|-1, 1\rangle + \sqrt{2}\hbar\,|0, 0\rangle\big) = 2\hbar\,|0, 0\rangle + \hbar\,|1, -1\rangle + \hbar\,|-1, 1\rangle\,.$$

Wait—let us be more careful. We have:

$$\hat{J}_{1-}\,|1, 0\rangle = \hbar\sqrt{2}\;|1, -1,\ 0\rangle \to \sqrt{2}\hbar\;|-1, 0\rangle\,,$$
$$\hat{J}_{2-}\,|1, 0\rangle = \hbar\sqrt{2}\;|0, -1\rangle \to \sqrt{2}\hbar\;|0, -1\rangle \text{ (but here } m_2 = 0).$$

Let us redo this properly. We write the full action:

$$\begin{aligned}(\hat{J}_{1-} + \hat{J}_{2-})\frac{1}{\sqrt{2}}\big(|1, 0\rangle + |0, 1\rangle\big) &= \frac{1}{\sqrt{2}}\Big[\sqrt{2}\hbar\,|0, 0\rangle + \hbar\,|1, -1\rangle + \hbar\,|-1, 1\rangle + \sqrt{2}\hbar\,|0, 0\rangle\Big] \\ &= \frac{\hbar}{\sqrt{2}}\big[2\sqrt{2}\;|0, 0\rangle + |1, -1\rangle + |-1, 1\rangle\big].\end{aligned}$$

Setting equal to $\hbar\sqrt{6}\;|2, 0\rangle$:

$$\boxed{|2, 0\rangle = \frac{1}{\sqrt{6}}\big(|1, -1\rangle + 2\,|0, 0\rangle + |-1, 1\rangle\big).}$$

Step 4: Orthogonality for $j = 1$.

In the $m = 1$ subspace, spanned by $\{|1,0\rangle\,,\ |0,1\rangle\}$, the state orthogonal to $|2,1\rangle$ is:

$$\boxed{|1,1\rangle = \frac{1}{\sqrt{2}}\big(|1,0\rangle - |0,1\rangle\big).}$$

Note the *antisymmetric* combination (versus the symmetric one for $j = 2$).

Step 5: Orthogonality for $j = 0$.

In the $m = 0$ subspace, spanned by $\{|1,-1\rangle\,,\ |0,0\rangle\,,\ |-1,1\rangle\}$, the state orthogonal to both $|2,0\rangle$ and $|1,0\rangle_{j=1}$ is:

$$\boxed{|0,0\rangle = \frac{1}{\sqrt{3}}\big(|1,-1\rangle - |0,0\rangle + |-1,1\rangle\big).}$$

Symmetry Pattern for $1 \otimes 1$

A beautiful result emerges:

- $j = 2$ states are **symmetric** under exchange $m_1 \leftrightarrow m_2$.
- $j = 1$ states are **antisymmetric** under exchange.
- $j = 0$ state is **symmetric** (with alternating signs).

In general, the highest-j multiplet is always symmetric. The symmetry then alternates: symmetric, antisymmetric, symmetric, ... as j decreases by one.
This pattern is crucial in particle physics (Griffiths): when combining identical bosons or applying the Pauli principle to fermions, only certain j values are allowed depending on the required overall symmetry of the state.

Connecting the Two Textbooks

Zettili's approach (Ch. 7) emphasizes the formal quantum-mechanical framework: he introduces the coupled and uncoupled bases carefully, derives the CG coefficients via the ladder operator technique, and presents applications in atomic physics (spin-orbit coupling, Zeeman effect, hyperfine structure). He also provides explicit CG tables.

Griffiths' approach (*Intro. to Elementary Particles*, Chs. 4–5) puts the same CG machinery to work in particle physics. Isospin is treated as a "flavor angular momentum," and the CG decomposition is used to:

- Classify mesons and baryons into $SU(2)$ multiplets,
- Predict branching ratios and cross-section ratios in strong decays,
- Understand why certain decay channels are forbidden.

The mathematical structure is identical—only the physical interpretation changes from "spin up/down" to "proton/neutron" (or "up quark / down quark").

Bottom line. The Clebsch–Gordan coefficients are a universal tool: learn them once in quantum mechanics, and you can apply them everywhere—from atomic spectroscopy to the quark model.